

$$T = \frac{2\pi}{\omega}, \quad \frac{m}{N_0} = \frac{\Omega}{2\pi}.$$

Chapter 1 – Signals

Sampling theorem:

$$F_s > 2f_{\max}.$$

Even and odd components:

$$x_{\text{even}}[n] = \frac{1}{2}(x[n] + x[-n]), \quad x_{\text{odd}}[n] = \frac{1}{2}(x[n] - x[-n]).$$

Angular frequency:

$$\omega = 2\pi f, \quad \omega = \Omega f_s.$$

Block processing: when N samples are available, the number M of windows with L samples and a shift of S samples is

$$M = \left\lfloor \frac{N-L}{S} \right\rfloor + 1.$$

Ideal sampling:

$$x_s(t) = \sum_n x[n]\delta(t - nT_s).$$

Power and energy:

$$P_{\text{sinusoid}} = \frac{A^2}{2}, \quad E_{e^{-at}u(t)} = \frac{1}{2a}, \quad a > 0.$$

$$P = \frac{E}{\Delta t}, \quad E = \int_{\langle \Delta t \rangle} p(t) dt.$$

$$A \cos(\omega t + \phi) \Rightarrow P = \frac{A^2}{2}, \quad x(t) = k \Rightarrow P = k^2.$$

SNR:

$$\text{SNR}_{\text{linear}} = \frac{P_x}{P_n}, \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \left(\frac{P_x}{P_n} \right).$$

For uniform quantization noise in the interval $[-\Delta/2, \Delta/2]$,

$$P_n = \frac{\Delta^2}{12}.$$

Autocorrelation:

$$R(\tau) = \frac{A^2}{2} \cos(\omega\tau), \quad R[i] = \frac{1}{2} \cos(\Omega i).$$

Unbiased autocorrelation of a signal $x[n]$ with length N :

$$R_x[\ell] = \frac{1}{N-|\ell|} \sum_n x[n+\ell]x^*[n], \quad \ell = -(N-1), \dots, N-1.$$

Unnormalized autocorrelation:

$$R_x[\ell] = \sum_n x[n+\ell]x^*[n].$$

Unnormalized cross-correlation:

$$R_{xy}[\ell] \triangleq \sum_{n=-\infty}^{\infty} x[n+\ell]y^*[n] = \sum_{n=-\infty}^{\infty} x[n]y^*[n-\ell].$$

A/D and D/A modeling:

$$x(t) \rightarrow \text{sampling} \rightarrow S/D \rightarrow \text{quantization}.$$

$$x_q[n] \rightarrow D/S(x_{s,q}(t)) \rightarrow \text{reconstruction} \rightarrow x(t).$$

Useful identities:

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}, \quad \sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}, \quad \Omega_k = \frac{2\pi k}{N}.$$

$$e^{j0} = 1, \quad e^{j\pi/2} = j, \quad e^{j\pi} = -1, \quad e^{j3\pi/2} = -j, \quad e^{j2\pi} = 1.$$

Chapter 2 – Transforms

One-sided Z-transform pair:

$$a^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{1}{1 - az^{-1}}, \quad \text{ROC} : |z| > |a|.$$

Orthonormal DFT pair:

$$X[k] = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x[n](W_N)^{nk}, \quad x[n] = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X[k](W_N)^{-nk}.$$

Forward and inverse DFT matrices for $N = 4$:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}, \quad A^{-1} = \frac{A^H}{N} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}.$$

Forward DCT matrix obtained with $A=\text{dctmtx}(3)$ in MATLAB:

$$A = \begin{bmatrix} 0.577 & 0.577 & 0.577 \\ 0.707 & 0.000 & -0.707 \\ 0.408 & -0.816 & 0.408 \end{bmatrix}.$$

This matrix is orthonormal, and its inverse is the Hermitian transpose.

Common Z-transform pairs:

Continuous-Time Fourier Transform (FT) pairs:

Residues in partial-fraction expansions:

$$A_i = (s - p_i)X(s)|_{s=p_i}.$$

Discrete-Time Fourier Transform (DTFT) pairs:

$$r = e^{-aT}, \quad c = \cos(\omega T), \quad s = \sin(\omega T).$$

Z-transform properties:

Summary of Fourier series and transforms: